EXP. NO: 26

MATLAB SIMULATION OF SINGLE-BIT ERROR DETECTION DATE: AND CORRECTION USING SYNDROME VECTORS IN LINEAR BLOCK CODES

Objectives

i) To simulate single-bit error detection and correction using syndrome vector decoding in a linear block coding scheme with MATLAB.

ii) To analyze the encoding, error detection, syndrome computation, and correction process by comparing the transmitted, received, and corrected codewords.

CODE

```
clc;
clear;
close all;

% (7,4) Hamming Code
msg = [1 0 1 1];

% Encoding
code = zeros(1,7);

% Place data bits
code(3) = msg(1);
code(5) = msg(2);
code(6) = msg(3);
code(7) = msg(4);

% Calculate parity bits (even parity)
code(1) = mod(code(3)+code(5)+code(7),2);
code(2) = mod(code(3)+code(6)+code(7),2);
code(4) = mod(code(5)+code(6)+code(7),2);

% Introduce single-bit error at position 3
rx = code;
rx(3) = 1 - rx(3);

% Error detection (syndrome calculation)
s1 = mod(rx(1)+rx(3)+rx(5)+rx(7),2);
s2 = mod(rx(2)+rx(3)+rx(6)+rx(7),2);
s4 = mod(rx(4)+rx(5)+rx(6)+rx(7),2);

error_pos = s1 + 2*s2 + 4*s4;
```

```
% Correct the error
corrected = rx;

if error_pos ~= 0
   corrected(error_pos) = 1 - corrected(error_pos);
end

% Extract corrected message
dec = corrected([3 5 6 7]);

% Display results
disp('Message Bits:');
disp(msg);

disp('Encoded Bits:');
disp(code);

disp('Received Bits with Error:');
disp(rx);

disp('Error Position:');
disp(error_pos);

disp('Corrected Codeword:');
disp(corrected);

disp('Corrected Message:');
disp(dec);

% Plot results
figure;

subplot(2,1,1);
stem(1:7, code, 'filled');
title('Encoded Hamming Codeword');
xlabel('Bit Position');
ylabel('Bit Value');
grid on;

subplot(2,1,2);
stem(1:7, rx, 'filled');
title('Received Codeword with Single-Bit Error');
xlabel('Bit Position');
```

```
ylabel('Bit Value');
grid on;
```

OUTPUT



